Spire 3D Viewer: Technical Project Report

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Abstract

Spire 3D Viewer is an Android app that lets a distributor, sales representative or engineering team scan a QR code on a Spire products datasheet and inspect the matching meter as a 3D model on a phone. The app was built by one developer between June and September 2026, after an earlier period of research and discussion with the company's decision makers. It uses Flutter, a bundled model-viewer page and 3D models stored in Firebase Storage, and its production release on Google Play is in early access as of October 2026. This report describes the problems, the users, the scope, the design, the code, and the current limits

1. Introduction and Problem

Spire Metering Technology LLC (Spire) is a small water meter manufacturer with 10 to 20 long-time employees that sells its products worldwide. Because its sales and technical support team is small, Spire relies on outside partners, including independent engineers, local engineers, distributors, and engineers hired by project management teams, to install its products. These partners are often far away, may speak different languages, and may be unfamiliar with Spire's meters. Explaining the product through phone calls, emails, photos, and specification sheets is time-consuming for a small team, and two-dimensional materials cannot fully show the meter's shape, parts or fit. Shipping physical samples overseas is costly and slow. At the same time, Spire competes with larger companies that have bigger sales teams. The Spire 3D Viewer Demo Version is a first step in exploring how these problems might be addressed. It presents an accurate, to-scale, interactive 3D model of one Spire water meter (280W-TAP) that partners and customers can view on their own devices with the aim of improving installation support and product presentation.

The idea began as a company web page showing 2D images. It changed into a scanner app with its own QR codes for two main reasons:

1) Large 3D models served from the company's public website would add load and traffic to the site, so the models are kept in separate cloud storage and opened only through the app.
2) The company wants the models seen mainly by distributors and engineers who hold a Spire products datasheet, not by anyone who finds a link

2. Target Users

Target user	Who they are	Need or problem	How the Spire 3D Viewer Demo Version can help
Distributors	Companies that resell Spire meters in their regions	Must learn and explain the product to their own customers without physical samples	View the meter's shape and details anytime, and use the model when presenting it to customers
Independent and local engineers	Engineers who design or oversee installations near the project site	Need to understand dimensions, connections, and fit before installation	Examine an accurate, to-scale model to plan installation and check fit in a meter pit or pipe run
Engineers hired by project management teams	Engineers working on specific utility or construction projects	Must evaluate and specify meters, often in formal bids	Inspect the product's design and dimensions during selection and bid evaluation
Installers and field technicians	Crews who physically install the meters	May be unfamiliar with Spire meters and have limited support on site	Use the model as a visual reference for orientation, parts, and connections
Utility buyers and procurement officers	Staff at water utilities who choose and purchase meters	Must compare products, often from larger, better-known brands	Explore the meter interactively instead of relying only on photos and spec sheets
Spire sales and technical support staff	Spire's own small team	Spend time repeatedly explaining the products; samples are costly to ship	Share one consistent reference with partners, reducing repeated explanations and sample shipments

The first release of the Spire 3D Viewer Demo Version, available as Early Access, covers one product: the 280W-TAP cap meter. The company plans to expand the viewer to seven models, one for each of its seven products datasheets, so that each datasheet's QR code opens the matching 3D model.

3. Objectives and Scope of Work

The demo release has one objective: scanning a Spire QR code should open an interactive 3D model of the product in a few taps.

In scope (demo release)

- Scan a QR code with the app's camera and verify that it is a Spire code.
- Download the 3D model (GLB file) from cloud storage, with a loading indicator.
- Rotate and zoom the model using touch gestures.
- Switch between Closed Cap and Open Cap views of the 280W-TAP cap meter.
- Close the viewer and return to the scanner.

Out of scope (future work)

- 3D models for the other six products and their datasheets.
- Datasheets and pipe-connection details inside the viewers.
- A hold-to-scan gesture on the camera screen.

Work completed

Period	Work
June 15 to July 10, 2026	3D modeling of the 280W-TAP meter (Open Cap and Closed Cap), including physical measurements of the original product using a digital caliper.
Late July 2026	Scanner and viewer app first built
August 6, 2026	Adjustments and testing
August to September 2026	Closed testing with 12 testers
September 20, 2026	Google Play production access granted
October 2026	Full rollout (open testing and early access) submitted; ready to release a public version

Andy Chen is the sole developer. He created the 3D model, the app, and the QR generator, and he also handled the release setup and the testing.

4. Project Description and System Design

The app performs four tasks in order: scan, verify, download and display. Figure 1 shows how the parts connect.

Only a verified Spire code leads to a model download

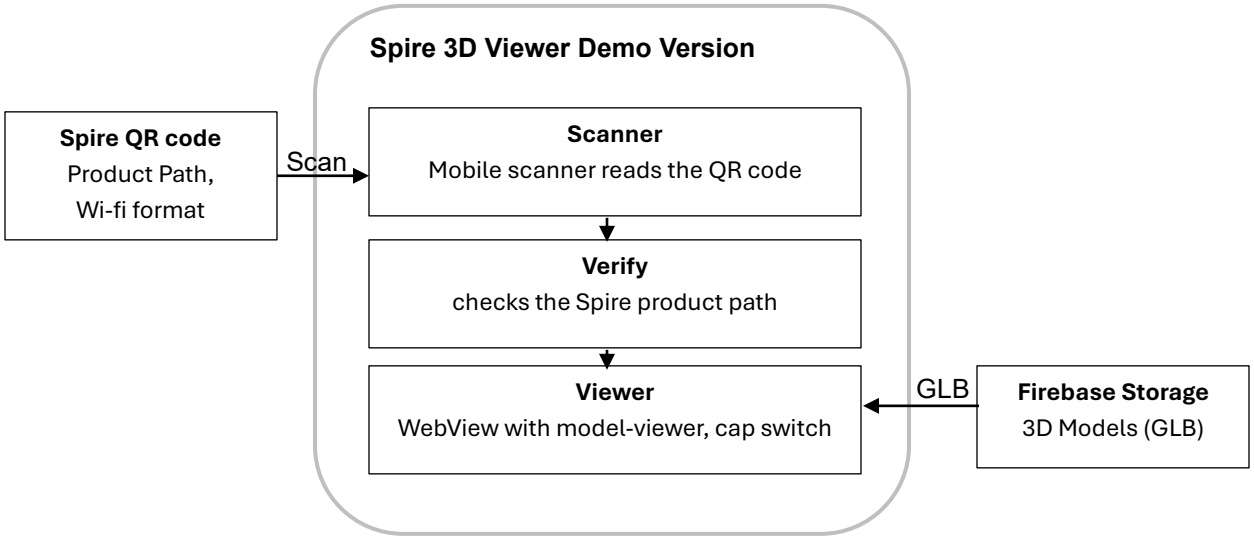


Figure 1. System architecture, showing its three parts

The app verifies the scanned code before it requests any model from cloud storage.

How a session works

1) The user downloads the Spire 3D Viewer Demo Version from Google Play and opens it.
2) The user points the camera at a Spire QR code on a product datasheet.
3) The app reads the code and checks that it contains the Spire path for a known product (for the demo, spire.com/meter/280w-tap)
4) If it matches, the screen shows “Meter Verified! Opening 3D Model...” and the app starts the download.
5) The 3D model (a GLB file) is fetched from Firebase Storage, and a progress percentage is shown.
6) A web view with Google’s <model-viewer> component displays the model. The user can rotate and zoom it, and switch between Closed Cap and Open Cap views.

Design Rationale

- **A custom scanner instead of a web link.** A QR code that opens a web page lets anyone with the link see the model. Here, the QR code holds a product

path wrapped in a Wi-Fi-style format, so a normal phone camera does not open it as a web page. Only Spire's scanner app treats it as a product code. This keeps causal viewers out, but it is not strong security (see Section 8).

- **Models downloaded at runtime.** The GLB files are kept out of the app package, which keeps the installed app small and allows the models to be updated without a new release.
- **One verified code per model.** Each of the seven planned products gets its own QR code, printed on its own datasheet. In the demo, the accepted product path is written into the app, so each new product currently requires an app update; keeping the product list in cloud storage would remove that step.
- **Cap switching in the viewer.** The page swaps between the Closed Cap and Open Cap models with a function call, so the user never leaves the viewer.

5. Implementation

The app is a Flutter (Dart) project that combines a camera scanner, a web view, and a small HTML page. The code is publicly available in the **GitHub repository**: https://github.com/AndySChen2027/Spire-3D-Viewer-Important-Code-Functions; in every sample below, storage addresses and access token have been replaced with placeholders.

Component	Tool	Job
Scanner	mobile_scanner (Flutter package)	Reads the QR code through the camera
Viewer	webview_flutter (Flutter package)	Hosts the bundled HTML page
3D display	Google <model-viewer>	Displays the GLB model and handles rotation and zoom
Model storage	Firebase Storage	Stores the Closed Cap and Open Cap GLB files
QR code generator	Python qrcode library	Creates the codes printed on product datasheets
Branding	flutter_launcher_icons, flutter_native_splash, Pillow	Creates the app icon, splash screen, and padded logo

Sample 1: QR code payload (Python). The QR code holds a product path in a Wi-Fi-style format rather than a web address. This sample is simplified from generate_qr.py.

```
payload = 'WIFI:S:;T:nopass;P:;H:true;;spire.com/meter/280w-tap'
qr = qrcode.QRCode(error_correction=qrcode.constants.ERROR_CORRECT_H)
qr.add_data(payload)
qr.make(fit=True)
qr.make_image().save('capmeter_280WTAP_invisible.png')
```

Sample 2: Verification (Dart). The scanner accepts a QR code only when it contains the known Spire product path. This sample is simplified from mainAppLogicAndUI.dart.

```
if (code.contains('spire.com/meter/280w-tap')) {
  // show "Meter Verified! Opening 3D Model..." and open the viewer
}
```

Sample 3: Cap switching (JavaScript). The bundled HTML page switches between the Closed Cap and Open Cap models with a single function call. This sample is simplified from webViewEntryCode.html.

```
function setCapMode(mode) {
  // mode is 'open' or 'closed'; point model-viewer at the matching GLB
  viewer.src = (mode === 'open') ? OPEN_CAP_URL : CLOSED_CAP_URL;
}
```

OPEN_CAP_URL and CLOSED_CAP_URL are placeholders for the storage links, which are not published here. A JavaScript channel named ModelViewerChannel tells the app when loading starts and finishes, so the app can show and hide the progress indicator.

6. 3D Model Samples, Testing and Results

The demo covers one product, the 280W-TAP cap meter, modeled in two sates (Closed Cap and Open Cap). Andy Chen built the model in Blender between June 15 and July 10, 2026, using digital caliper measurements of the physical meter, see Appendix C and Appendix D.

User flow. Appendix B shows the complete session on a phone, in order: the QR code on the SpireTap Series product datasheet; opening the app and aiming the camera at

the code; the “Meter Verified! Opening 3D Model….” screen; the loading screen with a process percentage; the Closed Cap and Open Cap models in the viewers; and closing the viewer to return to the scanner.

Testing. A 12-person closed test ran before production access was granted. A 39-second demo shows the main functions working on a real phone: rotating the Closed Cap model by finger, tapping Open Cap, a loading indicator rising through 50%, 66%, 81% and 88%, and the Open Cap model rotated to show the inside.

Figure 2. 39-second demo video on YouTube: https://youtu.be/L0MGT0P6rJg

Results so far

- The scan, verify, download and display process works end to end on a phone.
- Rotation, zoom and switching between Closed Cap and Open Cap work as designed.
- The Open Cap model took about 10 seconds to load in the demo recording.

7. Release on Google Play

Google granted production access on September 20, 2026. In October 2026, the full rollout, including open testing and early access, was submitted, and the app is available on Google Play as of this report. Google’s production-access email of September 20, 2026, is included in Appendix A. To find it, search for “Spire 3D Viewer” on Google Play.

8. Limitations and Future Work

- **Light access control.** A normal camera app does not open the QR code, and the app checks the product path, which discourages casual access. However, this is not strong security: the model links are stored inside the published app and could be extracted. Because the app is public on Google Play and the product datasheets are printed, anyone who has both can view the models. Stronger options would include signed or expiring links and server-side checks.
- **Load time.** Models are downloaded at runtime, so the first view depends on the network connection. Smaller GLB files, compression, and caching could shorten the load time.
- **Open product.** Only the 280W-TAP model has been released; models for the other six products are planned, and each will be linked to its own product datasheet.
- **Viewer content.** Datasheet information and pipe-connection details are planned additions to the viewer.
- **Scanning.** A hold-to-scan gesture is a planned improvement to the camera screen.

- **Android only.** The demo is released only on Google Play. Because Flutter can also build for iPhone, an iOS version is planned.

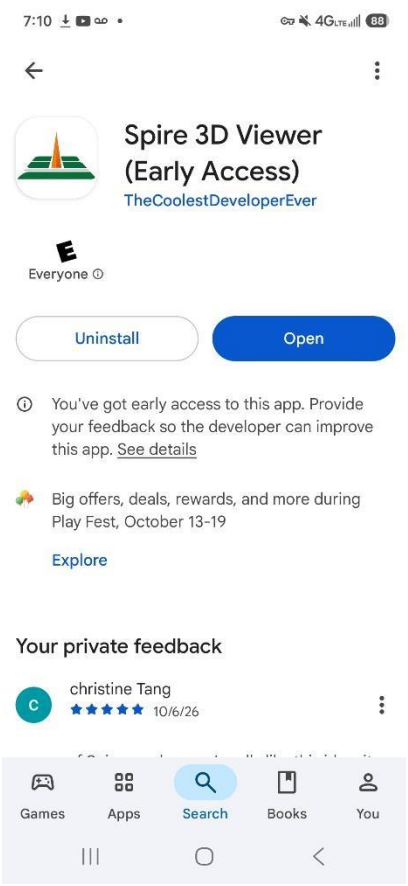
9. Conclusion

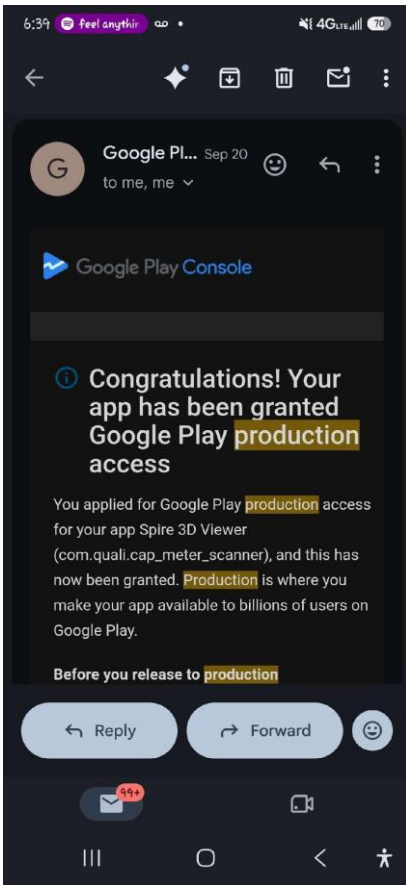
The Spire 3D Viewer Demo Version shows that one developer can turn 2D product information into an interactive 3D product view in a short period, from initial research to release on Google Play, using a small app, a bundled viewer page, and cloud-hosted models. The next steps are completing the remaining six models and measuring load times more systematically.

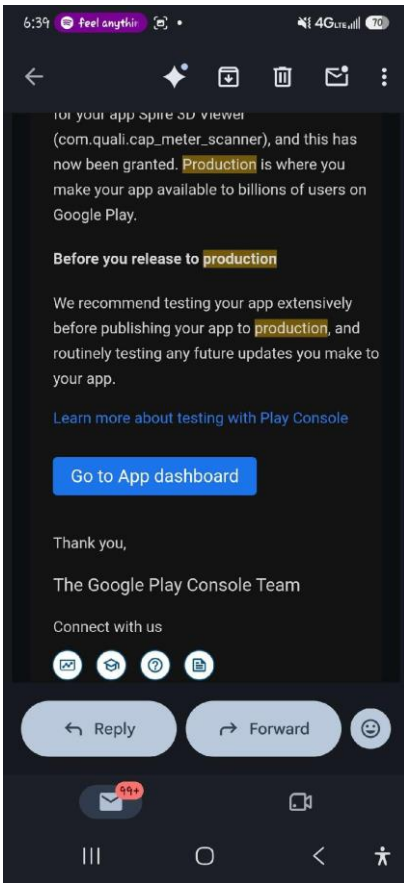
Disclosures

This report was developed with help from Andy's teacher, and Spire's sales and technical support team. Andy Chen reviewed the report and is responsible for its content.

Appendix A. Google Play production access







Appendix B. Spire 3D Viewer Demo Version: user flow



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Appendix C. Measurements of the Physical Meter Using a Digital Caliper









Appendix D. 3D Modeling in Blender

